

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-ae37a28d39b6ae91c.jsonl`
- SHA-256: `34a09378a2e7b5049515b0e75ff86f2c7ef74dda7248aa0f2c5a742f1e58d463`
- Source modified: `2026-06-12T14:44:52+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_774b13f4-334`
- session_id: `da34752f-c664-4262-9602-f5d4cfcc6db4`
```

Transcript

1. user (2026-06-12T14:40:53.934Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).

Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [medium] COGS figure cited to the wrong artifact ('2026-06-10 quality table') docs/PMF_FIELD_SIGNALS.md lines 30-31 attribute the serving-COGS figure '~\$0.30-0.60 per hour of video (~?25-50/hr)' to the '2026-06-10 quality table'. The 2026-06-10 quality table (docs/NEXT_STEPS.md, 'The quality table that now drives strategy') contains only IoU-vs-golden figures (0.657/0.744/0.83/0.93 etc.) ? no cost numbers. The \$0.30-0.60/hr figure actually comes from the 2026-06-10 web-grounded PMF audit, which today lives only in session memory (memory/current-state.md line 388: 'Gemini COGS ~\$0.30-0.60/hr'). The only in-repo cost figure is docs/COMMERCIALIZATION.md section 5 / C7: '~\$0.05 input for a 10-min clip' (= \$0.30/hr input-side only, which corroborates the low end but not the 0.60 ceiling). The number itself is plausible and the ? conversion is correct (~?85/USD gives ?25-51), but in a doc whose entire purpose is pre-registered honesty ? and under the project's attribution working agreement (never mis-state provenance) ? the citation label is wrong.

Proposed fix: Change the parenthetical to cite the real source, e.g. '(~?25-50/hr; 2026-06-10 web-grounded PMF audit; input-side corroborated by COMMERCIALIZATION.md section 5: ~\$0.05/10-min clip)'. Better: backport the \$0.30-0.60/hr COGS figure into COMMERCIALIZATION.md C7 or PMF_MARKET_RESEARCH.md so the repo-visible record carries it.

2. user (2026-06-12T14:41:02.681Z)

docs/pmf-field-signals

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md

M docs/NEXT_STEPS.md

?? arial.ttf

```
?? artifacts/
?? docs/FIELD_VISIT_ANSWER_SHEET.md
?? docs/PMF_FIELD_SIGNALS.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

3. user (2026-06-12T14:41:03.782Z)

```
1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9      [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12     first visit** ? what those answers *mean* and what totals trigger which decision.
13     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14     15 ambiguous conversations as whatever we hoped to hear.
15
16     ---
17
18     ## 1. The hypotheses under test (each maps to script questions)
19
20     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21     |---|---|---|---|
22     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
28     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31     hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32     roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
33     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37     Score each hypothesis on the answer sheet **from what they said and showed, not
38     what you inferred**. When between two grades, take the lower one.
39
40     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41     |---|---|---|---|
42     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
```

44 | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I review with students Mondays") | "could be useful, maybe" | no consumer identified |

45 | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it, satisfied, sticky |

46 | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free" |

47 | **H6 multi-court** | complains about it in their own words before/without prompting | agrees when asked | multi-court footage already works for them |

48 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

49

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral* signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

54

55 **Disconfirming-evidence quota:** every visit sheet must record at least one verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

59

60 **## 3. Pre-registered decision rules (n = 10?15 academies)**

61

62 Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED** ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

69 | **PIVOT A** ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

70 | **PIVOT B** ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C** ? payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

72 | **FALSIFIED** ? beachhead assumption fails** | <3 academies record at all (H1) **and** ?0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update

80 ``PMF_MARKET_RESEARCH.md`` §4 with field-verified rows (today's table is web-sourced).

82

83 **## 4. Mechanics**

84

85 - One [``FIELD_VISIT_ANSWER_SHEET.md``](FIELD_VISIT_ANSWER_SHEET.md) per academy, same day as the visit, in the associate's words + verbatim quotes.

87 - The owner (or Claude session) transcribes each sheet into one row of the rollout table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.

89

```

90 - Demo asset if a conversation earns it: the real compiled reel
91   (`output/MahadevpuraSingles_highlights.mp4`) on a phone ? *show, never promise*
92   (flyer ground rule 3 still applies; the associate doesn't carry the demo,
93   the owner does on follow-ups).
94
95   ### Rollup table (one row per completed visit)
96
97   | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98   |---|---|---|---|---|---|---|---|---|---|---|---|
99   | 1 | | | | | | | | | | |
100
101   ### Tally ? decision read-out (recompute per 5 visits)
102
103   | Metric | Count | Threshold in play |
104   |---|---|---|
105   | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106   | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107   | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108   | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109   | H1 any recording | /n | <3 ? FALSIFIED leg |
110   | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112   ## 5. Guardrails that bind the field work
113
114   - **Phase 1 records nothing** ? notes only (flyer rule 7).
115   - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116     players; **sessions with minors are never recorded for our library** (training-pool
117     guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118   - Answer sheets contain personal contact details ? they stay between the associate
119     and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:41:13.918Z)

```

1   # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3   > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4   > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5   > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7   ## Visit basics
8
9   | | |
10  |---|---|
11  | Date / time | |
12  | Academy name | |
13  | Area (neighbourhood) | |
14  | # courts in the hall | |
15  | Spoke to (name + role) | |
16  | Contact (phone/WhatsApp) ? *only if offered* | |
17  | How long the conversation ran | |
18
19  ## The seven questions ? their answers (verbatim where possible)
20
21  **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23  >
24
25  **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27  >
28
29  **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30

```

```

31 >
32
33 **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34 What's good, what's missing, what does it cost them?**
35
36 >
37
38 **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39 THEY say first (?)? What was their face doing?**
40
41 >
42
43 **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45 >
46
47 **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48 to call)**
49
50 >
51
52 ## Required: the most negative thing they said (verbatim)
53
54 >
55
56 ## Best quotes (anything vivid, in their words)
57
58 >
59
60 ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score
lower)
61
62 | Hypothesis | Score |
63 |---|---|
64 | H1 recording behavior | GREEN / AMBER / RED |
65 | H2 dead footage | GREEN / AMBER / RED |
66 | H3 audience | GREEN / AMBER / RED |
67 | H4 whitespace | GREEN / AMBER / RED |
68 | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69 | H6 multi-court pain | GREEN / AMBER / RED |
70 | H7 Phase-2 door | GREEN / AMBER / RED |
71
72 ## Competitor block (only if they named one)
73
74 | | |
75 |---|---|
76 | Product | |
77 | What it does well (their words) | |
78 | What it misses (their words) | |
79 | What they pay | |
80
81 ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
82
83 ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
84
85 ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
really decides)
86
87 >
88

```

5. user (2026-06-12T14:41:14.231Z)

```

1 # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2

```

3 ****Duration:**** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
academies are active)

4 ****Location:**** Bengaluru ? in-person visits to badminton academies

5 ****Compensation:**** ? ___ per completed academy visit + travel reimbursement *(discussed
individually)*

6 ****Reports to:**** Avi (phone/WhatsApp: ___)

7

8 ---

9

10 **##** The gig in one paragraph

11

12 We're building an AI tool that takes ordinary practice/match video from a badminton
academy and automatically finds

13 every rally, builds highlight reels, and generates match reports. Before building
further, we need honest, on-the-ground

14 answers from real academies about how they record, what they'd use, and what they'd pay.
****Your job:** visit 10?15 good

15 academies, have a friendly 15?20 minute conversation with the owner/head coach using the
script below, and write down

16 what they actually say.****** No selling. No tech knowledge needed. If a conversation goes
well, there's a Phase-2 follow-up

17 (coordinating a paid video-recording session) with additional compensation.

18

19 ---

20

21 **##** Phase 1 ? The conversation (what you actually do)

22

23 ****Who to talk to:**** the academy owner or head coach ? the person who decides what the
academy spends money on.

24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is
available.

25

26 ****Opening (keep it this honest):****

27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton
academies ? it takes normal practice

28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a
few serious academies to

29 > understand how you work with video today. Could I ask you a few questions? Takes 15
minutes, and we're not selling

30 > anything today."

31

32 ****The questions (ask in this order; the starred ones matter most):****

33

34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired
person?

35 2. ? ****Do you have recorded footage sitting around that nobody has time to watch?****
(multi-court hall recordings,

36 tournament days, CCTV) ? roughly how much?

37 3. Who would watch rally clips or highlights if they existed ? coaches for training?
players? parents?

38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory,
BadPro+, just YouTube/phone.)

39 What do you like / what's missing in it?

40 5. ? ****If you could hand over a full session's video and get back every rally cut out +
a highlight reel + a simple**

41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per
match?****** (Let THEM say a number

42 first. Note their face, not just the words.)

43 6. Your hall has multiple courts running at once, right? Does that make video less
useful today?

44 7. Would you let us record one session here (we pay a small fee for the time) to show
you what the tool produces on

45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

46

47 ****After each visit, fill ONE answer sheet ? the template is**

```
[`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)
48     (print one per visit):** academy name, location, # courts, person spoken to + role +
contact, answers 1?7 (their
49     words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a
50     Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51     [`PMF_FIELD_SIGNALS.md`](PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just
capture honestly.)*
52
53     ---
54
55     ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57     1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58         figure out ? what would feel fair to you?" Never quote a price yourself.
59     2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60         what happened to that file?" gets the truth. Prefer the second kind of question.
61     3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62         (a paid recording session + free sample output) and only as "would you be open to
it".
63     4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64         than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65     5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66         doesn't, and what they pay. Spend extra time here.
67     6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68         and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69     7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70         the academy or players unless the owner offers.
71     8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72     9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73         don't share the answer sheets with anyone except Avi.
74
75     ---
76
77     ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79     You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80         one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81         record 1?2 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82         written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83         our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84         the tool).
85
86     ---
87
88     ## Interested?
89
```

90 Message Avi at ____ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91 walkthrough) so you're never improvising.
92

6. user (2026-06-12T14:41:23.353Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-11-****human correction loop****. Every tester who accepts/rejects/nudges a rally boundary in the review
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-12-screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the per-user
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-13-training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-world,
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-14-multi-court-biased footage ? the distribution where the owned model's 0.66 ship target lives.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-15-docs\UI_PROPOSAL.md:16:Serving economics per the 2026-06-10 quality table: alpha serves the ****Gemini path****
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-17-(0.93 on clean footage, ~\$0.05/run) ? no GPU required in the request path. The WASB/owned-model
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-18-path stays on the owner's box and is ***not*** exposed in the alpha UI (segmenter choice hidden).
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-19-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-20-****The CLI is not replaced.**** The web app talks to the same FastAPI surface the CLI drives; the CLI
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_PROPOSAL.md-21-remains the power-user/ops tool.
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-26-| H5 | ****Willingness to pay per match**** at a price that clears unit economics | "should be free" / <?100 everywhere | Q5 ? |
docs\PMF_FIELD_SIGNALS.md-27-| H6 | ****Multi-court chaos is a felt problem**** (our moat: indexing messy BYO multi-court footage) | multi-court video already useful to them | Q6 |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-28-| H7 | ****Behavioral commitment**** ? they'll let us record on their court (Phase-2 door) | polite no across the board | Q7 |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-29-docs\PMF_FIELD_SIGNALS.md-30-Unit-economics floor behind H5: serving COGS on the Gemini path is ~\$0.30?0.60 per
docs\PMF_FIELD_SIGNALS.md:31:hour of video (? 225?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-32-roughly 75?150 to process, so ****?300/match** has real margin, ?100/match is
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-33-break-even-ish at retail****** ? that's why the script's anchor ladder is ?100/300/500.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-34-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-35-## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-36-
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-18-eval_baselines/heuristic_lovo_n6.json --version <semver>) and produced artifact v0.1.0 (LOVO ****0.626****, floor passed,
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-19-sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The ****n=6** owned corpus is in the collector******
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-20-(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-21-battery is done****** (see table) ? the moat is quantified.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-22-docs\NEXT_STEPS.md:23:## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-24-| Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-25-|---|---|---|
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-26-| Heuristic (velocity windowing) | 0.657 | 0.157 |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-27-| **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md-28-| Two-stage WASB+Gemini refine | 0.83 | ? |

7. user (2026-06-12T14:41:23.870Z)

docs\BACKLOG.md:35:- ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the cap. **Revisit:** before bumping supervision past 0.30.
docs\DEFERRED_HARDENING_PLAN.md:26: (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).
docs\DEFERRED_HARDENING_PLAN.md:129:- **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
docs\COMMERCIALIZATION.md:48:| C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
docs\COMMERCIALIZATION.md:64:| **`supervision`** (ByteTrack) | MIT | No | ? LIVE | Pinned `<0.30.0` ? migrate before bump (C2). |
docs\COMMERCIALIZATION.md:122:- ** (C) Gemini 2.5 Flash (paid):** input \$0.30/1M tokens, output \$2.50/1M; video ~263
docs\PMF_FIELD_SIGNALS.md:30:Unit-economics floor behind H5: serving COGS on the Gemini path is ~\$0.30?0.60 per
docs\PMF_FIELD_SIGNALS.md:31:hour of video (? 25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
docs\archives\MONETIZATION_AUDIT.md:79:- ** (C) Gemini 2.5 Flash (paid):** input **\$0.30 / 1M tokens** (text/image/video), output **\$2.50 / 1M** . Video tokenizes at **~263 tokens/sec** at default resolution ? a 10-min clip ? 158k input tokens ? **~\$0.05 input** (+ small output). Per-video API cost is **cheap** ; the economic risk is *volume × latency* , not unit price. Flash-Lite (\$0.10/\$0.40 per 1M) is a cheaper tier; Pro is ~10× pricier.
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:172: 0.751, doubles 0.737, Mahadevpura 0.657, but Kushagra **0.157** / gBaaddy **0.309**).

8. user (2026-06-12T14:41:34.636Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened** :
4      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
5      `docs/UI_ALPHA_EXECUTION_PLAN.md` ; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87** ;
6      **PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.
14
15     ## Where we are (one paragraph)
16     ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)
17     behind a CI-enforced import wall; the featurizer is single-sourced
18     (`backend/features/rally_seq.py`, train==serve);
```

```

17     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
18     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
19     sign test 5/6 wins  $p=0.109$  ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
20     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
21     battery is done** (see table) ? the moat is quantified.
22
23     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
24     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
25     |---|---|---|
26     | Heuristic (velocity windowing) | 0.657 | 0.157 |
27     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
28     | Two-stage WASB+Gemini refine | 0.83 | ? |
29     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
30
31     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
32     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
33     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
34     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
35     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
36     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
37
38     ## Prioritized next steps
39     1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
40     Priority order for new videos: **a) multi-court badminton** (the target distribution
AND where the ship target
41     lives), **b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
42     corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
43     `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
44     label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
45     The paired sign test needs n: at n=10 with 9 wins,  $p>0.011$  ? significance is
reachable this month.
46     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
47     reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
48     (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
49     paired-sign  $p<0.05$  vs heuristic" ? owner to ratify/adjust.
50     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51     2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52     calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54     badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55     by C3 (provenance multiplies per sport).
56     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:

```

```

57      "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
58      (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
59      `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
60      physical-visit approach):** `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (recruit hand-
out: script, ground rules,
61      Phase-2 consent; owner fills ?/contact placeholders) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture
62      form) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R scoring anchors + **pre-registered
decision rules** ? PROCEED /
63      PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
64      6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
65      deletion in AI Studio); **rename repo off "badminton"** (owner action,
66      in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
67
68      ## Pointers / artifacts
69      - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
70      Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
71      `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
72      - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
73      manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
74      - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
75      (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
76      labels: `...\Collect\Golden Labelled\`.
77      - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
78      (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
79      with a MISLEADING "credits depleted" 429; ramp gradually.
80      - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
81      PLATFORM slice = async job model (single-tenant).
82

```

9. user (2026-06-12T14:41:36.445Z)

```

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:330: never tie-inflated; fps-fix mainly helped the 30fps short
fold 0.30?0.56). Heuristic LOVO **0.480** unchanged (scored
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:384: in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ?
headline TT-0.909/pickleball-0.308 evidence is one cleanup from
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:388: subs rev $2.75M +12% YoY; PB Vision $99/yr + **$8/hr-of-
video partner API**; Gemini COGS ~$0.30?0.60/hr ? fat margin);
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:430: gBaaddy 0.309/testlarge 0.267/mahadevpura 0.657 ? exact).
**Contrast metric recomputed independently from raw CSVs:**
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:466: **WASB transfers to TT remarkably well!**); **PICKLEBALL
best F1 0.308** (poor ? wiffle ball too out-of-distribution).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:1336:- **baseline** thresholds (0.02,2.0,2.0): precision 0.577,
recall 0.600, **F1 0.588**, meanIoU 0.765.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\monetization-research-status.md:24:**A** eliminate via offline ActionFormer/MS-
TCN (MIT, Phase-4) . **B** self-host Qwen2.5-VL-7B/32B (Apache) . **C** keep Gemini paid-tier.
Rough per-video cost A <$0.05 . B $0.05?0.30 . C $0.05?0.40.

```

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md:49:(0.308) ? per-sport detectors or Gen-2.

10. user (2026-06-12T14:41:52.672Z)

375 rejects the plan's own MIT ActionFormer/ASFormer heads); (2) **Gemma-4 still
"approved"** in those same docs vs
376 AMBER/bench-only in the plan; (3) **M0.3 purge pointer WRONG** ?
`training.label\_candidates` is a pure GT-agnostic fn;
377 the live Gemini-silver training violation = `backend/eval/distill\_local.py` (+
calibrate_local thresholds) + gitignored
378 `output/local\_rally\_model.json`; (4) **C3 escape hatch broken** ? "train-own weights
(Phase-4 roadmap)" cites a phase
379 that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T
train a detector) ? own-weights is
380 an unscoped new workstream (coordinate labels or detector swap); (5)
OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are
381 linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS
is branch-only; (6) README
382 "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default";
(7) ROADMAP still says audio
383 "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve";
(8) Roora multisport CSVs live ONLY
384 in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline
TT-0.909/pickleball-0.308 evidence is one cleanup from
385 loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per
sport"); (10) ship-gate F1@IoU still
386 undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no
tracker (BACKLOG lacks it).
387 - **PMF (web-grounded, sources in audit output):** pricing/cost benchmarks VERIFIED
(SwingVision \$179.99/yr ~20k paying
388 subs rev \$2.75M +12% YoY; PB Vision \$99/yr + **\$8/hr-of-video partner API**; Gemini
COGS ~\$0.30?0.60/hr ? fat margin);
389 but ***"India badminton whitespace" FALSIFIED ? Bangalore is the most contested
badminton-AI city** : Machaxi (\$1.5M
390 Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School),
Game Theory (Gopichand-backed smart
391 courts w/ auto-highlights), Stupa (\$3.3M, BWF-approved, TT?badminton) ? NONE in our
2026-06-07 competitive table.
392 Surviving wedge (narrow, real): **software-only rally-precise indexing of EXISTING
messy multi-court BYO footage**
393 (market solves it only with per-court hardware) sold per-hour as engine/API, possibly
THROUGH those players
394 (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) +
**Gemini-wrapper baseline vs the
395 n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid
academy pilot.
396 - **AUDIT NOW COMPLETE (8/8 auditors, resumed `wa4d2980l` after Max-20x refresh; 40
agents).** New high/critical findings
397 beyond the first batch:
398 - ? **CRITICAL (collector) ? LICENSING GUARDRAIL VIOLATION:** `src/cli/ingest.py:12`
defaults `--license MIT`; the
399 ShuttleSet **annotation** license is stamped onto **YouTube broadcast video** (e.g.
shuttleSet_21 = a YONEX Thailand
400 Open broadcast, `is\_commercial=true, license\_type=MIT`). **24 videos / 1915 segments
currently exported as
401 "commercially-safe" on unsound grounds** ? directly violates the "training data must
be commercial-clean" guardrail.
402 - **HIGH (indexer, all verified):** (a) **nested config sections silently drop unknown
keys** ? only `AppConfig` has
403 `extra='allow'`, so the ENTIRE `indexing.wasb` block
(weights/conda_env/velocity_threshold) + `yolo\_prefetch\_workers`
404 + `debug\_duration` are unconfigurable via config.json; meanwhile
`yolo\_parallel\_chunks/yolo\_chunk\_workers` are read
405 nowhere. (b) **detector artifacts keyed by file BASENAME not video_id** ? two

```
`match.mp4` files reuse stale frames ?
406         a trajectory + candidate_segments computed from the WRONG video (silent training-
data poisoning). (c) **re-ingest is
407         destroy-before-confirm** ? `add_video('processed'/'failed')` INSERT-OR-REPLACE
cascade-deletes prior good segments
408         BEFORE the new run; a failed/empty re-run leaves zero. (d) **stitching padding
config not wired** ? API reels always
409         end_padding=1.0 despite config 0.5. (e) **`/api/process` arbitrary server-file read
+ `/api/compile` path-traversal/
```

11. user (2026-06-12T14:41:53.504Z)

No matches found

12. user (2026-06-12T14:42:03.456Z)

```
-rw-r--r-- 1 avidu 197609 17273 Jun 10 14:14 C:/Users/avidu/Projects/badminton-highlight-
indexer/docs/COMMERCIALIZATION.md
-rw-r--r-- 1 avidu 197609 16871 Jun 8 07:11 C:/Users/avidu/Projects/badminton-highlight-
indexer/docs/PMF_MARKET_RESEARCH.md
```

13. user (2026-06-12T14:42:04.716Z)

```
134:| **Veo (Cam 3)** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; $115M raised (2025) |
139:| **SwingVision** | **Tennis/pickleball AI, single iPhone** | Software freemium | **USD
179.99/yr**; free 2 hrs/mo | Closest analog; 200k+ monthly users; $10M+ raised (2025) |
```

14. user (2026-06-12T14:42:11.202Z)

```
1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30     counsel . ? future
31     Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33     order.)
34
35     31
```

```

32      | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33      |---|---|---|---|---|---|
34      | | **? PENDING (needs action / not started)** | | | |
35      | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See $4. Decision: provision Ada-class GPU? |
36      | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37      | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
`DEFERRED_HARDENING_PLAN.md` (#16). |
38      | | **? IN-FLIGHT (in progress / watch)** | | | |
39      | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` $5A. |
40      | C13 | **Beachhead validation** | **P0** | ? | Market research
(`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
$6. |
41      | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42      | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | **Ship-gating** blocker (multiplies per
sport). Resolve via an own-trained detector **or** a clean MIT/Apache detector swap ? NOTE:
ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-
weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |
43      | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned
video-QA model must build on a **commercial-clean (MIT/Apache-2.0/BSD), video-capable** base ?
primary **Qwen3-VL (Apache-2.0)**, fallback **InternVL3 (MIT)**. **Gemma 4 = AMBER / bench-
only** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build
target yet). **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision
(naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM
(Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented
dataset + eval harness**, not the model. | **MIT/Apache-2.0/BSD** for code+data+weights; **open-
model teachers only**; separate explicit **training opt-in**; **exclude minors**; per-clip
consent ledger. See `OWNED_MODEL_IMPLEMENTATION_PLAN.md` (authoritative). |
44      | C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC**
uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain
residual; confirm decode-side exposure with counsel before launch. |
45      | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini
(~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline
costs unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing.
See $5. |
46      | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL/JSON/CSV + `basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-
player/match stats, clip gallery. See $6. |
47      | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports
surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more
`VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
48      | C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation
ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
49      | | **? COMPLETED** | | | |
50      | C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8
replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL
code/weights in the stack. | Done. Keep new deps permissive. |
51

```

```

52      ---
53
54      ## 2. Licensing & dependency posture (refreshed risk table)
55
56      Verdicts updated against the current stack (`requirements.txt` + code). Original
57      2026-05-30 verdicts and the 25-claim verification transcript are in
58      `archives/MONETIZATION_AUDIT.md`.
59
60      | Dependency | License | SaaS obligation? | Current verdict | Notes |
61      |---|---|---|---|---|
62      | `~ultralitics` + `yolov8n.pt`~~ | AGPL-3.0 | Yes ($13) | ? **REMOVED** | Replaced per
63      ADR-005 ? no longer in the stack. |
64      | **torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen
65      replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`). |
66      | **supervision** (ByteTrack) | MIT | No | ? LIVE | Pinned `<0.30.0` ? migrate before
67      bump (C2). |
68      | **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open
69      (C3). |
70      | **MonoTrack / YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
71      | **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH |
72      Must bill the paid tier to keep user video private (C4). |
73      | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? |
74      Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |
75      | ffmpeg/ffprobe, google-genai, torch, opencv, numpy,
76      fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as
77      subprocess / imported; all permissive. |
78      | **obsreport** (reports) | permissive (private) | No | ? soft dep | Pin the
79      `git+ssh@vX` tag once the repo is published (BACKLOG). |
80
81      **Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)
82      are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
83      and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
84
85      ---
86
87      ## 3. Runtime AI strategy ? A/B/C
88
89      The semantic rally-boundary step is the one external/metered dependency. Three paths
90      (from the audit), with current standing:
91
92      - ***(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
93      classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
94      cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
95      shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
96      - ***(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
97      per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
98      - ***(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
99      ongoing metered cost + external dependency. **?* current live path.**
100
101      Hosted alternatives for (C) comparison: Anthropic Claude (no training on
102      inputs/outputs; customer owns outputs; **no native video** ? weaker here);
103      OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
104
105      ---
106
107      ## 4. Codec / output economics (the live gap ? C5)
108
109      **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
110      codecs it implements. Royalties attach to the act of encode/decode.
111
112      - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
113      is first 100k units/yr royalty-free with a permanent "free to end-users over the
114      internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
115      - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
116      (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing

```

```

108     must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110     highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - ** (C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - ** (B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - ** (A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:
142 1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143 players,
144 formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
145 who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
146 not
147 the individual; needs INR-tuned pricing.
148 2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8%
149 YoY)
150 but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
151 head-to-head, not greenfield.
152 3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China
153 needs
154 local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
155
156 Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B
157 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixelot/
158 Spiideo; comparable to SwingVision software-only).
159
160 ### PMF risks (from the readiness review)
161 - **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
162 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
163 narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
164 - **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;
165 per-rally tags, stats, and a correction loop are what make it sticky (C10).
166 - **Golden labels are the recurring blocker** across the offline segmenter, audio
167 fusion, and measurable quality ? the highest-leverage investment (C8).
168
169 ### Refined priority order (quality/data phase ? "focus on quality once new videos
170 land")
171 Carried from the readiness review; commercialization items above are tracked in
172 parallel.

```

166

167 1. ****Golden-set expansion + data-flywheel UX**** (highest with new labeled videos):

168 ingest the new batch, expand calibration/CV/regression baselines, make label

169 contribution + safe retraining easy and non-contaminating (eval/training separation).

170 2. ****Hybrid unification + observability hooks**** (deferred item 7, now higher): extract

171 shared logic from `tracknet\_hybrid`/`wasb\_hybrid` before heavy AI timing/metrics

land.

172 3. ****Detector/windowing quality iteration**** with new data: re-tune velocity/merge/

173 min-duration + `rally\_gate`; validate WASB on real video; trial audio fusion

(ADR-007).

174 4. ****Deeper value layer**** (beyond minimal exports, C10): per-rally tagging/notes ?

175 `candidate\_segments`, per-player/match stats, clip gallery, human-correction loop.

176 5. ****Input-quality enforcement & guidance polish**** (C11): more guideline dimensions ?

177 validators/report rules; "your video is good enough" UX.

178 6. ****Multi-sport maturation + recording best practices.****

179 7. ****SaaS / multi-user / async foundations**** (C12) ? lower until monetization is firm.

180 8. ****Polish & distribution**** (C9): installer story, first-run path, codec/output,

economics.

181

182 ****Recommendation:**** with the mid-week labeled-video batch, spend ~80% on #1 + #3

183 (flywheel + quality iteration on real data), then unification (#2) and deeper value

(#4).

184

185 ---

186

187 ## 7. Open questions needing owner / counsel sign-off ??

188

189 1. ****Ultralytics Enterprise price**** ? only relevant if a permissive detector ever proves

190 insufficient (fallback). Quote-only.

191 2. ****Per-checkpoint / dataset licenses**** for WASB / TrackNetV3 / any VLM ? train-own or

192 confirm each (C3).

193 3. ****Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds**** ? confirm live

194 figures with counsel before relying on the "small SaaS under threshold" reading

(C5/C6).

195 4. ****Real per-video GPU cost & latency**** of self-hosted VLM vs Gemini paid at expected

196 throughput (C7).

197 5. ****Provision Ada-class GPU (L4/L40)**** for HW AV1 encode? (C5 ? gates the codec

switch.)

198

199 ---

200

201 ## Changelog

202

203 - ****2026-06-07**** ? Added the market-research-of-record (`PMF\_MARKET\_RESEARCH.md`):

204 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13

205 "beachhead validation" workstream. Wired the summary into S6.

206 - ****2026-06-07**** ? Created this living tracker. Merged the 2026-05-30 monetization

207 audit + the commercial-readiness review. ****Refreshed since the audit:**** C1 AGPL

208 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,

209 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to

210 `DEFERRED\_HARDENING\_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer

211 exports). Archived `COMMERCIAL\_READINESS\_REVIEW.md`.

212

213 ---

214

215 ## See also

216 - `PMF\_MARKET\_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,

beachhead; cited).

217 - `archives/MONETIZATION\_AUDIT.md` ? licensing/economics research-of-record (sources +

verification transcript).

218 - `archives/COMMERCIAL\_READINESS\_REVIEW.md` ? superseded assessment (history).

219 - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor

process).

220 - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).

221 - `DEFERRED\_HARDENING\_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).

222 - `GOLDEN\_SET\_VENDORING.md` ? vendor process + Roora pilot (C8).
223 - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
224

15. user (2026-06-12T14:42:48.423Z)

187: (P .81 R .84, 32 preds) . **full-video wrapper 0.93** (P 1.0 R .87).
191:- **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops the wrapper to 0.66 = the
204:- ? **WRAPPER BASELINE (gemini-flash-latest, ~\$0.2-0.5 total): Mahadevpura F1 0.93** (P 1.00 R 0.87, mIoU 0.80, ±0.7s;
215:- **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) + owned model's job = multi-court
216: (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of the wrapper (provider-fallback
235:waves. ?? commit/PR the 503-backoff WIP . machine-side 4K re-run (chunks should be ~tens of MB) . task-17 battery
260: battery on unblock (in task 17): wrapper Mahadevpura ? score; wrapper Kushagra (multi-court MOAT test); two-stage

16. user (2026-06-12T14:42:49.496Z)

No matches found

17. user (2026-06-12T14:43:30.058Z)

180 **Checkpoint:** 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was fine, postpay active; ALL
181 failures were burst/capacity artifacts). Merged this stretch: **#80** (chunker re-encode/downscale, from the chip
182 session ? reviewed+merged), **#81** (rescued the chip branch's orphaned run_signals test via cherry-pick), **#83**
183 (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), **#84** (tracked
184 `eval\_baselines/gemini\_wrapper\_2026-06-10.json`). Suite 336 green.
185 - **FINAL QUALITY TABLE (IoU .5 vs human golden labels):**
186 - **Mahadevpura** (clean-ish, 31 rallies): heuristic 0.657 . Gen-0 0.744 . **two-stage WASB+Gemini refine 0.83**
187 (P .81 R .84, 32 preds) . **full-video wrapper 0.93** (P 1.0 R .87).
188 - **Kushagra** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 . Gen-0 0.561 .
189 **wrapper 0.66** (P .69 R .62, ±1.2s; FPs = background-game pickups + one 53s dead-time merge ? exactly the
190 contrast-metric confound).
191 - **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops the wrapper to 0.66 = the
192 owned model's ship target** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth reach). Two-stage refine quality
193 (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only candidate clips uploaded).
194 Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback architecture is load-bearing.
195 - Artifacts: `output/mahadevpura\_gemini\_refined.csv`,
196 `output/MahadevpuraSingles\_highlights.mp4` (demo reel),
197 Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~\$1-2.
197 - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court target now has a number: 0.66);
198 C13 interviews + camera pilot; repo rename.
199
200 **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory CORRECT). New key in gitignored
201 `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE artifact: text ping + serial
202 upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai\_max\_workers`

```

(was hardcoded 5; config ships
203     1=serial ? at 5 workers the same video 429'd, at 1 it worked).
204     - ? **WRAPPER BASELINE (gemini-flash-latest, ~$0.2-0.5 total): Mahadevpura F1 0.93** (P
1.00 R 0.87, mIoU 0.80, ±0.7s;
205     3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected)
vs Gen-0 learned 0.744 vs
206     heuristic 0.657. **Clean footage = commoditized.**
207     - ?? **MOAT TEST (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56** (P .54 R
.58 mIoU .69) ? Gen-0 learned
208     0.561, both >> heuristic 0.157. **The multi-court confound hurts Gemini too ? the
wedge survives on the target
209     footage class.** Caveats: only chunks 2+4 answered (5/7 lost to GENUINE gemini-flash-
latest capacity 503s ? retry
210     off-peak for full coverage); visual-only (audio stripped in downscale); 1280p re-
encode (4K chunks were 1.3GB >
211     500MB upload cap ? the audit's silent-oversize-skip fired live; fix = chip
task_44e72c41 ? see PR #80 checkpoint
212     below). Kushagra preds keyed under downscaled-file MD5 (`output/kushagra_1280.mp4`).
213     - ? **DEMO ARTIFACT: `output/MahadevpuraSingles_highlights.mp4`** ? real 27-rally reel
compiled by VideoCompiler
214     (the once-broken module) from Gemini segments. The academy-pitch object exists.
215     - **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) +
owned model's job = multi-court
216     (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of
the wrapper (provider-fallback
217     story strengthens). PENDING: gemini_refine two-stage (503-gated, run off-peak), full
Kushagra re-run, PR #80 merge.
218
219     **Checkpoint:** 2026-06-10 (4K gemini oversize fix ? **MERGED: #80 + #81 + #82; master
`f01387e`, CI rollout SUCCESS**).
220     Root cause of the live "0 segments / success" on KushagraYashAviFirstVideo.MP4 (4K):
gemini segmenter stream-copies 90s
221     chunks ? all 7 at 832?1299MB > provider `MAX_UPLOAD_MB=500` ? every chunk silently
skipped. Fix (3 layers): (1) chunks
222     RE-ENCODED + downscaled to new declared knob `indexing.ai_chunk_scale_width` (default
1280 = gemini_refine's
223     `--clip-scale-width`; 0 = legacy stream-copy; also fixes keyframe drift); (2) skip is
LOUD ? provider
224     `metrics["oversize_skipped"]` + segmenter aggregation (chunked + single-file) ?
`logger.error` + NEW thread-safe

```

18. user (2026-06-12T14:43:31.676Z)

```

docs\NEXT_STEPS.md:24:| Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
docs\NEXT_STEPS.md:59:  `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE
(2026-06-12, owner ratified the
docs\PMF_FIELD_SIGNALS.md:91:  (`output/MahadevpuraSingles_highlights.mp4`) on a phone ? *show,
never promise*
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:110:6 distinct matches labeled (3 singles incl.
Mahadevpura, 1 doubles, 2 multi-court); per-match + LOVO recorded; the
docs\archives\quality-iterations\README.md:20:[Omitted long matching line]
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:133:| Mahadevpura (singles,
+n=6) | 0.67 | 74% | 33% | **2.2x** | borderline (some background) |
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:137:The contrast metric
predicts heuristic F1 **monotonically across 5 matches** (3.9x?0.73 ? 1.4x?0.16) ? Mahadevpura's
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:172:  0.751, doubles 0.737,
Mahadevpura 0.657, but Kushagra **0.157** / gBaaddy **0.309**).
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:173:- **n=6 added
MahadevpuraSingles** (borderline multi-court, contrast 2.2x): held out and trained on the *other
5*, the

```

19. user (2026-06-12T14:44:45.649Z)

Structured output provided successfully

20. assistant (2026-06-12T14:44:52.085Z)

Verification complete. The claim is real: the citation at docs/PMF_FIELD_SIGNALS.md:30-31 attributes the \$0.30?0.60/hr COGS figure to the "2026-06-10 quality table," which contains only IoU figures; the actual source is the 2026-06-10 web-grounded PMF audit living only in session memory (current-state.md:388). Severity assessed as low rather than medium ? the number is correct, partially corroborated in-repo, no decision rule changes, and the fix is a one-line citation correction (optionally plus backporting the figure into COMMERCIALIZATION C7).